

Systemic Treatments: Biologics, Biosimilars, and Oral Treatments



patient **navigation**
center



Introduction

This booklet is designed to give an overview of how systemic treatments work to treat psoriasis and psoriatic arthritis, the different types of systemic treatments, their benefits, and the possible risks and side effects. We hope this resource helps you to learn about these treatment options and make an informed treatment decision with your health care provider.

This booklet is based on guidelines developed by the American Academy of Dermatology (AAD) and the National Psoriasis Foundation (NPF) in the articles, "Joint AAD—NPF guidelines of care for the management and treatment of psoriasis with biologics" and "Joint American Academy of Dermatology—National Psoriasis Foundation guidelines of care for the management of psoriasis with systemic nonbiologic therapies." These guidelines were published as sections in the Journal of the American Academy of Dermatology in April 2019 through June 2020, respectively. It also includes a summary of the prescribing information for each biologic, biosimilar, and oral systemic treatment approved to treat psoriatic disease.

This version was published in December 2025 and will be reviewed and updated yearly. This resource is for educational use only and should not replace the care or advice of your health care provider. If you have any specific questions about how a systemic treatment may manage your psoriatic disease, please talk with your provider.



Psoriatic Disease

Psoriatic disease refers to psoriasis and psoriatic arthritis, as well as the related comorbidities and underlying systemic inflammation. A comorbidity is a disease or condition that is related to a health condition you have, such as psoriasis. While not everyone will develop psoriatic arthritis, about 1 in 3 people living with psoriasis will also have this comorbidity.¹ These 2 conditions may be experienced at the same time or at different points in a lifetime. Typically, people develop signs of psoriasis first, but in rare cases signs of psoriatic arthritis develop first.² Many people experience other comorbidities related to the systemic inflammation as well, including cardiovascular (heart) disease, diabetes, anxiety and depression, or inflammatory bowel disease (Crohn's disease or ulcerative colitis). To learn more about comorbidities download a copy of our Comorbidities Quick Guide from the NPF Patient Navigation Center at psoriasis.org/comorbidities-quick-guide/.

Psoriasis (sore-EYE-ah-sis)

Psoriasis is a chronic (recurring and long-term) disease related to the immune system. More than 7.5 million adults in the United States³, and 125 million people worldwide live with psoriasis according to the World Psoriasis Day consortium. Signs and symptoms can start at any age. Psoriasis is slightly more common in women than men, however, people of all ages, genders, races, and ethnicities may live with psoriasis.⁴

When you have psoriasis, your immune system is overactive and causes inflammation in your body. This inflammation can speed up skin cell growth. Normally, it takes 21-28 days for cells on the surface of the skin to grow and shed; it may take only 4 days in skin affected by psoriasis.⁵ This results in itchy or painful, scaly, inflamed plaques (patches) on your skin. Plaques may look red, pink, violet, or brown depending on how dark your skin is normally.



Psoriatic arthritis

Psoriatic arthritis is also a chronic disease related to the immune system. About 1 in 3 people living with psoriasis also develop psoriatic arthritis.¹ It can start at any age, but often appears between ages 30 and 50. Psoriatic arthritis may cause swelling, pain, and stiffness in your joints and in areas where your tendons and ligaments connect to bone (entheses). Pain in this location is named enthesitis.

While it is less common, people can develop psoriatic arthritis without having identified signs of psoriasis.² Some individuals have psoriasis for many years before developing any signs of psoriatic arthritis. This is why it is important to have your health care providers regularly screen you for psoriatic arthritis.⁶ If you think you might have psoriatic arthritis, you can take a short 5 question screener at psoriasis.org/psoriatic-arthritis-screening-test/. Share the results with your health care provider.

Treatments

Although there is not yet a cure, there are many ways to treat and manage psoriatic disease. Over the past 20 years, new treatments have been approved for psoriasis and psoriatic arthritis. Today there are more effective ways to manage psoriatic disease than ever before—and the results can be life-changing.

Your health care provider will recommend a treatment based on:

- Whether you have psoriasis, psoriatic arthritis, or both
- Whether your psoriatic disease is mild, moderate, or severe
- Your medical history and overall health
- Your experience with other treatments
- Your preference for how treatments are taken and how often you need to take them

Talk to your health care provider about any concerns of risks and side effects that you may have about a recommended treatment. It is important to find a treatment that you feel comfortable using.

When we talk about **risk**, we are talking about the chance you may develop a specific health issue. A risk is not a guarantee, only a possibility. Talk with your health care provider about any concerns you have about potential risks and steps you can take to lower your risk.

A **side effect** is a reaction to a medical treatment which may be undesirable to some patients. Your health care provider or pharmacist can tell you about common side effects. There may be ways to prevent side effects such as eating a meal before taking a medication to prevent nausea. Always talk to your health care provider about any concerns you have about side effects.

There are 4 types of prescribed treatments recommended to treat psoriatic disease: topical treatments, phototherapy, oral systemic treatments, and biologic treatments. You can learn about each treatment type at psoriasis.org/treatment-and-care. This booklet takes a closer look at oral systemic, biologic, and biosimilar treatments (together called systemic treatments).

Systemic treatments

Systemic treatments are prescription medications that work throughout the body. They are usually used for people with moderate-to-severe psoriasis but may be used in other cases depending upon the location of the plaques and how it affects your day-to-day life. Systemic treatments are also given to patients who cannot use or did not see results with topical treatment or phototherapy. Systemic treatments can be taken by mouth, given as an injection (shot) or as an intravenous (IV) infusion (a method of delivering fluids, including medication, into your bloodstream).

There are 2 types of systemic treatments:

- Biologics and biosimilars⁷
- Oral treatments⁸

Biologics and oral systemic treatments differ in how they address inflammation in the body. Biologics target specific parts of the immune system that play a role in psoriatic disease. Some newer oral treatments also act similarly by being more targeted. Meanwhile, traditional oral treatments may act on your whole body. Here is an analogy to describe how these types of treatment work:



Biologics, biosimilars, and some newer oral treatments

Putting out the fire with a firehose is like a biologic, biosimilar, or newer oral treatment blocking a specific pathway or element in the immune system.



Traditional oral treatments

Putting out the fire by dumping water from a helicopter is like a traditional oral treatment affecting the immune system more widely than biologics or some newer oral treatments.

Biologic and biosimilar treatments

Biologic and biosimilar treatments (also called biologics and biosimilars) are medications made from living cells that are given as an injection (shot) or as an intravenous (IV) infusion (slow drip of medication into the vein). Biologics target specific parts of the immune system that play a role in psoriatic disease. They decrease substances in the immune system that are responsible for inflammation and increased cell turnover in the skin and joints.

Some biologics are also considered disease-modifying antirheumatic drugs (DMARDs), which means they attempt to slow or stop the processes in the body that cause joint damage.

Some biologics are approved to treat children living with psoriasis and psoriatic arthritis. Some studies show that biologics work well in treating pustular and erythrodermic psoriasis in children, but this needs to be tested more in clinical trials.⁹ The U.S. Food and Drug Administration (FDA) has approved some biologics for children; a list of approved biologics can be found starting on page 10 of this booklet.

Possible risks

Anyone considering taking a biologic should talk with their health care provider about short- and long-term side effects and risks. It is important to weigh the risks and benefits of using the treatment.

Biologics can increase the risk of infection. If you develop any sign of an infection, such as a fever, cough, or flu-like symptoms, you should contact your health care provider right away.

Signs of infection include:

- Chills
- Damp, sticky feeling, or sweating
- Fever
- Nasal (nose) or chest congestion
- Pain or burning when urinating
- Shortness of breath
- Skin redness, swelling, soreness, or warmth



There is limited information on the impact of biologics on developing fetuses or breastfed babies.⁶ Talk to your health care provider if you are trying to become pregnant, are pregnant, or are breastfeeding to find a biologic that is safe for you and your baby.

Possible side effects

Each biologic has its own set of possible side effects, but here are the common side effects of taking a biologic¹⁰:

- Upper respiratory tract infection
- Skin reaction where the biologic is injected
- Flu-like symptoms
- Urinary tract infection
- Headache

People who should not use biologics

While each biologic has its own list of people who should avoid the treatment (listed on the paperwork that comes with your prescription), in general biologics may not be safe if you have⁷:

- A lowered immune system.
- Active tuberculosis (TB) or a positive TB test that has not been treated.
- An active infection.
- Recently received a “live” vaccine such as Zostavax for shingles, MMR (measles, mumps, and rubella), or nasal spray flu vaccine. While it is still important for you to receive vaccines, your health care provider may recommend you get an “inactivated” version of a vaccine which does not have live parts of the virus in the vaccine. For guidance about COVID-19 vaccines and psoriatic disease, visit the NPF COVID-19 Resource Center at psoriasis.org/covid-19-resource-center.

You may still be able to use biologics if any of the listed above apply to you. Talk with your health care provider about whether they may be appropriate. Your provider knows your medical history and can make the best recommendation.

Using biologics with other treatments

Your health care provider may prescribe a biologic alone or to be used with other treatments such as phototherapy, topicals, or oral treatments to treat your psoriatic disease.

If you are using a treatment (for another health condition) that suppresses your immune system, talk with your health care provider about the risks and benefits of using biologics to treat your psoriatic disease.

Types of biologics

Biologics are categorized according to what they target and inhibit (or block) in the body:

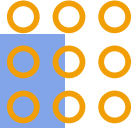
Type of biologic (mechanism of action)	Brand name
Tumor necrosis factor-alpha (TNF-α) inhibitors	Cimzia, Enbrel, Humira, Remicade, Simponi, and Simponi Aria
Interleukin-12 and 23 (IL-12/23) inhibitors	Stelara
Interleukin-17 (IL-17) inhibitors	Cosentyx, Siliq*, and Taltz
Interleukin-17A and 17F (IL-17A/17F) inhibitors	BIMZELX
Interleukin-23 (IL-23) inhibitors	Ilumya, Skyrizi, and Tremfya
Interleukin-36R (IL-36R) inhibitors	SPEVIGO
T-cell inhibitors	Orencia

*Siliq is an IL-17 receptor blocker

Tumor necrosis factor-alpha (TNF-α) inhibitors

TNF-α is a cytokine (a protein) that prompts the body to create inflammation. In psoriasis and psoriatic arthritis, lots of TNF-α is made in the skin and joints. That leads to the rapid growth of skin cells and damage to joint tissue. Blocking TNF-α production helps stop the inflammatory cycle of psoriatic disease.

Talk with your health care provider if you have any concerns about the potential risks or side effects of any treatment.



TNF-α inhibitors approved for psoriatic disease

• Cimzia • Enbrel • Humira	• Remicade • Simponi • Simponi Aria
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Avoid if you have⁸

- An allergy to TNF-α inhibitors
- Untreated hepatitis B infection
- A history of lymphoreticular malignancy (tumors in the lymphoid system)
- An active infection (including tuberculosis)
- Congestive heart failure
- Multiple sclerosis (MS) or a first-degree relative (parent, sibling, or child) with MS

Pregnancy and lactation⁷

- Pregnant women can take TNF-α inhibitors.
- TNF-α inhibitors (except for Cimzia) do affect the baby's immune system, so babies are considered immunosuppressed for at least 1 to 3 months after delivery.
- TNF-α inhibitors are safe for women who are breastfeeding.
- TNF-α inhibitors are safe for men who are trying to conceive with their partners.

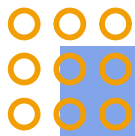
Potential risks include⁶

- Potential to increase the risk of infection
- Increase the risk of skin cancer and lymphoma

Interleukin-12 and 23 (IL-12/23) inhibitors

IL-12/23 inhibitors work by selectively targeting the proteins: interleukin 12 (IL-12) and interleukin 23 (IL-23). Interleukins 12 and 23 are associated with psoriatic inflammation.

Talk with your health care provider if you have any concerns about the potential risks or side effects of any treatment.



IL-12/23 inhibitors approved for psoriatic disease

- Stelara

Avoid if you have⁷

- An allergy to IL-12/23 inhibitors
- Untreated hepatitis B infection
- A history of lymphoreticular malignancy (tumors in the lymphoid system)
- An active infection (including tuberculosis)

Pregnancy and lactation⁶

- The safety of using IL-12/23 inhibitors during pregnancy and while breast-feeding is uncertain.
- IL-12/23 inhibitors may be prescribed if there is a medical need.
- IL-12/23 inhibitors are safe for men trying to conceive with their partners.

Interleukin-17 (IL-17) inhibitors

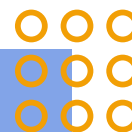
IL-17 inhibitors hinder IL-17 signaling, which interrupts the inflammatory cycle of psoriasis. The way in which these inhibitors work can vary. Two IL-17 inhibitors, Cosentyx and Taltz, hinder a cytokine called IL-17, which is involved in the inflammatory and immune responses. Another biologic, Siliq, is like Cosentyx and Taltz but works a bit differently. Siliq is an IL-17 receptor blocker, which prevents the start of inflammatory and immune responses. We encourage you to talk to your health care provider if you have any concerns about the potential risks or side effects of any treatment.

IL-17 inhibitors are not recommended to be used if you have an inflammatory bowel disease (IBD) such as Crohn's disease or ulcerative colitis. Talk with your health care provider before starting IL-17 treatment, especially if you or a family member has a history of IBD.

If you are prescribed Siliq (brodalumab), you may notice it has a warning on its label. This is a result of a small number of patients who had suicidal thoughts and behavior, including death by suicide, when they took Siliq (brodalumab) during clinical trials. It is not known if Siliq (brodalumab) caused the suicidal thoughts or behavior.¹¹ Talk to your health care provider if you have thought about suicide or had suicidal behavior recently or in the past. For your safety, your provider may prescribe a different IL-17 or biologic treatment.

Regardless of the treatment that you are taking, if you experience any thoughts of harming yourself or others, contact the Suicide and Crisis Lifeline at 988 or via chat at 988lifeline.org. Also, contact your health care provider to discuss if you should stop your treatment.

Talk with your health care provider if you have any concerns about the potential risks or side effects of any treatment.



IL-17 inhibitors approved for psoriatic disease

- Cosentyx
- Siliq
- Taltz

Avoid if you have⁸

- An allergy to IL-17 inhibitors
- A history of or active inflammatory bowel disease (IBD)
- Current thoughts of suicide or a history of suicidal ideation [especially for patients taking Siliq (also called brodalumab)]

Pregnancy and lactation⁷

- The safety of using IL-17 inhibitors during pregnancy and while breastfeeding is uncertain.
- IL-17 inhibitors may be prescribed if there is a medical need.
- IL-17 inhibitors are likely safe for men to take when trying to conceive with their partners.

Interleukin-17A and 17F (IL-17A/17F) inhibitors

IL-17A/17F inhibitors work by selectively targeting the proteins: interleukin-17A (IL-17A) and interleukin-17F (IL-17F). Interleukins 17A and 17F are associated with inflammatory responses.

Talk with your health care provider if you have any concerns about the potential risks or side effects of any treatment.

IL-17A/17F inhibitors approved for psoriatic disease

- BIMZELX

Avoid if you have¹²

- An allergy to IL-17A/17F inhibitors
- Active tuberculosis (TB)
- A history of or active inflammatory bowel disease (IBD)

Pregnancy and lactation¹²

- The safety of using IL-17A/17F inhibitors during pregnancy and while breastfeeding is uncertain.
- IL-17A/17F inhibitors may be prescribed if there is a medical need.



Interleukin-23 (IL-23) inhibitors

IL-23 inhibitors target IL-23, a cytokine that is linked with inflammation in psoriasis and psoriatic arthritis. IL-23 inhibitors work to reduce psoriatic symptoms and slow disease progression.

Talk to your health care provider if you have any concerns about the potential risks or side effects of any treatment.

IL-23 inhibitors approved for psoriatic disease

- Ilumya
- Skyrizi
- Tremfya

Avoid if you have⁷

- An allergy to IL-23 inhibitors

Pregnancy and lactation⁶

- The safety of using IL-23 inhibitors during pregnancy and while breastfeeding is unknown.
- IL-23 inhibitors may be prescribed if there is a medical need.



Interleukin 36 receptor (IL-36R) inhibitor

An IL-36R inhibitor works by inhibiting the activation of the interleukin 36 receptor (IL-36R). The interleukin 36 signaling pathway, which is part of the immune system, has been shown to be involved in causing generalized pustular psoriasis flares.¹³

Talk with your health care provider if you have any concerns about the potential risks or side effects of any treatment.

IL-36R inhibitors approved for psoriatic disease

- SPEVIGO

Avoid if you have¹³

- An allergy to IL-36R inhibitors
- Recently received or are scheduled to receive a live vaccine
- An active tuberculosis (TB) infection or have been in close contact with someone who has TB
- An active, serious infection



Pregnancy and lactation¹³

- IL-36R inhibitors should not be used during pregnancy or while breastfeeding

T-cell inhibitors

T-cells are a type of white blood cell that is involved in the inflammation in psoriatic disease. Biologics that inhibit T-cells keep them from becoming activated which results in a reduction in inflammation. Unlike targeted biologics, T-cell inhibitors more widely suppress the immune system.

Talk with your health care provider if you have any concerns about the potential risks or side effects of any treatment.

T-cell inhibitors approved for psoriatic disease

- Orencia

Avoid if you have¹⁴

- An allergy to T-cell inhibitors

Pregnancy and lactation⁶

- T-cell inhibitors may be prescribed if there is a medical need.¹⁴
- T-cell inhibitors are likely safe for men to take when trying to conceive with their partners.¹⁵



Biologics approved for psoriatic disease

The FDA has approved certain biologics to treat adults living with psoriasis, psoriatic arthritis, or both.

Brand name (scientific name)	Approved for Psoriasis	Approved for Psoriatic Arthritis
BIMZELX (bimekizumab-bkzx)	✓	✓
Cimzia (certolizumab pegol)	✓	✓
Cosentyx (secukinumab)	✓	✓
Enbrel (etanercept)	✓	✓
Humira (adalimumab)	✓	✓
Ilumya (tildrakizumab-asmn)	✓	
Orencia (abatacept)		✓
Remicade (infliximab)	✓	✓
Siliq (brodalumab)	✓	
Simponi (golimumab)		✓
Simponi Aria (golimumab)		✓
Skyrizi (risankizumab)	✓	✓
SPEVIGO (spesolimab-sbzo)	✓ *	
Stelara (ustekinumab)	✓	✓
Taltz (ixekizumab)	✓	✓
Tremfya (guselkumab)	✓	✓

*Approved to treat pustular psoriasis flares

The FDA has also approved the use of the following treatments among children and adolescents:

- **Enbrel** is approved to treat children 4-17 years old with moderate-to-severe plaque psoriasis (*Approved in 2016*) and children 2 years and older with juvenile psoriatic arthritis (*Approved in 2023*).
- **Orencia** is approved to treat children 2 years and older with psoriatic arthritis (*Approved in 2023*).
- **Stelara** is approved to treat children 6 years and older with moderate-to-severe plaque psoriasis (*Approved in 2020*) and children 6 years and older with active psoriatic arthritis (*Approved in 2022*).
- **Taltz** is approved to treat children 6 years and older with moderate-to-severe plaque psoriasis (*Approved in 2020*).
- **Simponi Aria** is approved to treat children 2 years and older with active psoriatic arthritis (*Approved in 2020*).
- **Cosentyx** is approved to treat children 6 years and older with moderate-to-severe plaque psoriasis (*Approved in 2021*) and children 2 years and older with active psoriatic arthritis (*Approved in 2021*).
- **SPEVIGO** is approved to treat children 12 years and older with generalized pustular psoriasis (GPP) flares (*Approved in 2024*).
- **Tremfya** is approved to treat children 6 years and older who weigh at least 88 pounds with moderate to severe plaque psoriasis (*Approved in 2025*) and children 6 years and older who weigh at least 88 pounds with active psoriatic arthritis (*Approved in 2025*).



Biosimilars

Biosimilar medications are a type of biologic drug and may be administered via injections (shots) or intravenous (IV) infusion. Biosimilars are very similar to a biologic treatment that is already approved by the FDA—which is often called the “reference product.” Keep in mind that biosimilars are not exact copies of their biologic reference products. Biologics are large and complex molecules from living sources that cannot be exactly copied.

Although biosimilars are modeled after an FDA-approved biologic, they undergo a different approval process. The FDA can approve a treatment as a biosimilar or an interchangeable biosimilar medication. FDA standards ensure that biosimilars and interchangeable biosimilar medications are just as safe and effective as their biologic reference product.

An interchangeable biosimilar medication must meet the biosimilar standard plus an additional standard that the treatment will produce the same clinical results as the reference product in any given patient. If a biosimilar is approved as interchangeable, in some states a pharmacist may substitute it without letting the prescribing health care provider know.

There are several biosimilars approved for the treatment of psoriasis and psoriatic arthritis, although they may not be available due to a variety of reasons:

Biologic the reference product	Biosimilar Brand Name (scientific name)
Humira (adalimumab)	Abrilada (adalimumab-afzb), AMJEVITA (adalimumab-atto), CYLTEZO (adalimumab-adbm), HADLIMA (adalimumab-bwwd), HULIO (adalimumab-fkjp), Hyrimoz (adalimumab-adaz) , Idacio (adalimumab-aacf), Simlandi (adalimumab-ryvk), Yuflyma (adalimumab-aaty), and YUSIMRY (adalimumab-aqvh)
Enbrel (etanercept)	Erelzi (etanercept-szsz) and Eticovo (etanercept-ykro)
Remicade (infliximab)	Avsola (infliximab-axxq), Inflectra (infliximab-dyyb), Ixifi (infliximab-qbtx), and Renflexis (infliximab-abda)
Stelara (ustekinumab)	Imuldosa (ustekinumab-srlf), Otulfi (ustekinumab-aaaz), Pyzchiva (ustekinumab-ttwe), Selarsdi (ustekinumab-aekn), Starjemza (ustekinumab-hmny), Steqeyma (ustekinumab-stba), Wezlana (ustekinumab-auub), Yesintek (ustekinumab-kfce)

People should not use biosimilars if:

- Your immune system is significantly compromised
- You have an active infection

Screening for tuberculosis or other infectious diseases is required before starting treatment with all biologics, including biosimilars.

Risks and side effects

The risks and side effects of biosimilars are the same as those associated with their biologic reference product. Anyone considering taking a biosimilar should talk with their health care provider about the short- and long-term side effects and risks. It is important to weigh the risks against the benefits.

When you are on a biologic or biosimilar, you may have a higher risk of infection. If you develop any signs of infection, contact your health care provider right away. For a list of signs of infections, please go to page 3 of this booklet.

Using biosimilars with other treatments

It is important to tell your health care provider about all treatments, medications, vitamins, or supplements that you are taking.

Like all biologics, biosimilars can be used with other treatment options including topicals and phototherapy. For example, the biologics Enbrel, Humira, Stelara, and Remicade are shown to be safe and effective when taken with methotrexate. This means that their biosimilars, including Erelzi, Amjevita, Wezlana, and Inflectra, may be safe and effective when taken with methotrexate.

If a treatment is not safe to take with a certain biologic, that treatment is not safe to take with the biosimilar for that biologic.

Talk with your health care provider to see if a biosimilar is a good treatment option for you.

Oral systemic treatments

Oral systemic treatments (also called systemic non-biologics) are medications you take by mouth. Different oral treatments work in different ways to decrease symptoms of psoriatic disease. Traditional oral treatments may act on your whole body. However, some newer oral treatments are similar to biologic treatments by being more targeted.

Looking again at the analogy comparing how biologics and oral systemics work, you can see that traditional oral treatments take a wider approach to managing a problem.



Biologics, biosimilars, and some newer oral treatments

Putting out the fire with a firehose is like a biologic, biosimilar, or newer oral treatment blocking a specific pathway or element in the immune system.



Traditional oral treatments

Putting out the fire by dumping water from a helicopter is like a traditional oral treatment affecting the immune system more widely than biologics or some newer oral treatments.

Some treatments, such as methotrexate, which is an oral systemic, are also disease-modifying antirheumatic drugs (DMARDs). DMARDs may relieve more severe symptoms, slow down joint and tissue damage, and stop the progression of certain types of psoriatic arthritis. For example, methotrexate does not treat axial (spine) psoriatic arthritis. Talk with your health care provider if you have any questions about DMARDs that may treat your psoriatic arthritis.

Oral systemic treatments may be prescribed for children with severe psoriasis who have not responded well to other treatments such as topical treatments and phototherapy. In some cases, health care providers prescribe oral systemic treatments for children to be used short-term until symptoms improve and then taper to the lowest dose and then off the oral systemic treatment completely.⁹

Possible risks

Some oral systemic treatments may suppress your immune system, causing greater risk of infection or organ damage. If you notice any sign of infection, contact your health care provider. For a list of signs of infections, please go to page 3 of this booklet.

You should not use certain oral systemic treatments, such as methotrexate or Soriatane (also known by its generic name, acitretin), if you are pregnant or breastfeeding.⁷ You may use another oral systemic treatment if there is a medical need during that time. Your health care provider can recommend what is safest for you and your baby.

Talk with your health care provider about whether oral systemics are right for you. Your provider knows your medical history and can make the best recommendation.

Using oral systemic treatments with other treatments

Oral systemic treatments can be used with other psoriasis treatments such as topicals or phototherapy. Some may also be used with biologics. Talk with your health care provider about other treatments you are using to treat your psoriatic disease or other health conditions.

Oral systemic treatments approved for psoriatic disease

Brand name (scientific name)	Approved for Psoriasis	Approved for Psoriatic Arthritis
Cyclosporine	✓	
Methotrexate	✓	*
Otezla (apremilast)	✓	
RINVOQ (upadacitinib)		
Soriatane (acitretin)	✓	
Sotyktu (deucravacitinib)	✓	
Xeljanz and Xeljanz XR (tofacitinib)		

*Methotrexate is used "off-label" to treat psoriatic arthritis, but it is only approved by the FDA to treat psoriasis.⁷

Cyclosporine

Cyclosporine is an immunosuppressive drug that was first used to help prevent rejection in organ transplant patients. In 1997, the FDA approved cyclosporine for adults with severe psoriasis and otherwise normal immune systems. Cyclosporine works by suppressing the immune system and slowing down the growth of certain immune cells.⁸

Cyclosporine is generally tolerated well in treating children with psoriasis. Cyclosporine can help to quickly control severe plaque, pustular, and erythrodermic psoriasis in children.⁹

You take cyclosporine by mouth daily as a pill or liquid. The liquid medication should be diluted by mixing it with room temperature water or fruit juice, except grapefruit juice.¹⁶ Cyclosporine can give fast relief from psoriasis symptoms. Improvements may happen in as little as 2 weeks or 3 to 4 months. Treatment with cyclosporine is **not** recommended for more than 1 year.⁸

Talk with your health care provider if you have any concerns about the potential risks or side effects of any treatment.

Avoid if you¹⁶

- Have a compromised immune system
- Have abnormal kidney function
- Have high blood pressure
- Have cancer, or a history of cancer (other than basal or squamous cell skin cancers)
- Have severe gout
- Are pregnant or breastfeeding
- Are undergoing radiation treatment
- Have received PUVA treatment (especially more than 200 sessions)

Common side effects include¹⁶

- Decreased kidney function
- Headache
- High blood pressure
- High cholesterol
- Excessive hair growth
- Tingling or burning sensation in the arms or legs
- Skin sensitivity
- Increased growth of gum tissues
- Flu-like symptoms
- Upset stomach
- Tiredness
- Muscle, bone, or joint pain

Potential risks include¹⁶

- Skin cancer if you have previously used methotrexate or other immunosuppressive agents, PUVA, UVB, coal tar, or radiation therapy
- Kidney damage
- Hypertension

May not be safe to use with¹⁶

- Certain antibiotics
- Anti-inflammatory drugs
- Anti-fungals
- Gastrointestinal agents
- Calcium channel blockers
- Anti-convulsants
- Certain OTC medications such as aspirin or ibuprofen
- Grapefruit and grapefruit juice
- Certain supplements
- Other prescription drugs including oral contraceptives



Methotrexate

The FDA approved the use of methotrexate in 1972 and has been used for nearly 5 decades to treat psoriasis in adults⁸ and moderate-to-severe psoriasis in children.⁹ For people with psoriasis, methotrexate binds to and inhibits an enzyme involved in the rapid growth of skin cells and slows down the cell's growth rate.

Methotrexate can come as a tablet to take by mouth or as an injection into your skin. Your health care provider will tell you how often you should take methotrexate. The schedule depends on the condition you have and on how your body responds to the treatment.

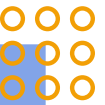
Your health care provider may also prescribe a folic acid supplement since you may experience lower levels of folate in your body when you start taking methotrexate. Taking a folic acid supplement will help to keep folate levels normal.

Take methotrexate exactly as directed. Do not take more or less of it or take it more often than prescribed by your provider.

Pregnancy should be avoided if either partner is taking methotrexate. Men should be off methotrexate for at least 3 months before trying to conceive. Women should wait at least 4 months after stopping methotrexate to try to become pregnant. In addition, women who are breastfeeding should not take methotrexate.

Your health care provider will recommend that you have regular blood tests to make sure your body is safely processing methotrexate.⁸

Talk with your health care provider if you have any concerns about the potential risks or side effects of any treatment.



Avoid if you¹⁷

- Are allergic to methotrexate or any ingredients in methotrexate
- Are an alcoholic or have alcoholic liver disease, cirrhosis, or other chronic liver diseases
- Have an immunodeficiency syndrome or an active infectious disease
- Are trying to get pregnant (applies to both men and women), pregnant, or breastfeeding
- Have an active peptic ulcer
- Have significant liver or kidney abnormalities
- Have underdeveloped bone marrow, a low white blood cell count, low platelets, or significant anemia

Common side effects include¹⁷

- Mouth sores
- Low white blood cells
- Nausea, upset stomach
- Feeling poorly
- Tiredness, chills, fever, dizziness
- Higher chance for getting infections
- Diarrhea
- Vomiting
- Hair loss
- Easy bruising

Potential risks from long term use include¹⁷

- Serious anemia, lower white cells, red cells, and platelets in your blood
- Liver damage
- Kidney damage
- Lung disease
- Cancer of the lymph system (lymphoma)

May not be safe to use with¹⁷

- Non-steroidal anti-inflammatory drugs (NSAIDs)
- Some antibiotics (such as pencillin)

Otezla (apremilast)

Otezla was approved in 2014 to treat moderate-to-severe plaque psoriasis and psoriatic arthritis. In 2021 it was approved to treat mild-to-moderate plaque psoriasis. In 2024, it was also approved to treat moderate to severe plaque psoriasis in children 6 years and older. This tablet, taken by mouth, treats psoriatic disease by regulating inflammation within the cell. Otezla inhibits an enzyme known as phosphodiesterase 4 (PDE 4). This enzyme controls much of the inflammatory action within cells, which can affect the level of inflammation associated with psoriatic disease. Otezla can reduce flaking, scaling, joint tenderness, and swelling.

There is limited research on the impact of taking Otezla on fertility, pregnancy, and breastfeeding. Your health care provider may recommend taking Otezla during these times if the benefits of treating your psoriatic disease outweigh the potential risks to you and your baby.⁸

Talk with your health care provider if you have any concerns about the potential risks or side effects of any treatment.

Avoid if you¹⁸

- Are allergic or sensitive to Otezla or its ingredients

Common side effects include¹⁸

- Diarrhea
- Nausea
- Vomiting
- Upper respiratory infection
- Runny nose, sneezing, or congestion
- Abdominal pain
- Tension headaches
- Headache

Potential risks from long term use include¹⁸

- Gastrointestinal problems, including severe diarrhea, nausea, and vomiting
- Depression
- Weight decrease

May not be safe to use with¹⁸

- Certain prescriptions
- Certain over-the-counter (OTC) medications



RINVOQ (upadacitinib)

RINVOQ was approved in 2021 to treat adults living with psoriatic arthritis. Taken once daily by mouth, RINVOQ works by targeting a specific part of the immune system (Janus kinase) to reduce the overactive immune response associated with psoriatic arthritis.¹⁹

There is limited information about the impact of taking RINVOQ on fertility, pregnancy, and breastfeeding. However, based on animal studies taking RINVOQ while pregnant may cause harm to a fetus. It is recommended women do not take RINVOQ during pregnancy and use effective contraception during the time they are taking RINVOQ and for 4 weeks after completing their treatment.¹⁹

Talk with your health care provider if you have any concerns about the potential risks or side effects of any treatment.

Avoid if you¹⁹

- Are allergic to RINVOQ or any ingredients in the medication
- Are pregnant or plan on becoming pregnant
- Are breastfeeding or plan to breastfeed
- Have recently received or are scheduled to receive a live vaccine
- Have tuberculosis (TB) or have been in close contact with someone with TB
- Have an active, serious infection

Common side effects include¹⁹

- Nausea
- Upper respiratory infection
- Cough
- Fever
- Acne

Potential risks from long term use include¹⁹

- Serious infections
- Cancer
- Major cardiovascular events such as a heart attack or stroke
- Blood clots
- Tears in the stomach or intestines

May not be safe to use with¹⁹

- Certain prescriptions including JAK inhibitors, biologic DMARDs, or with potent immunosuppressants such as azathioprine and cyclosporine
- Certain over-the-counter (OTC) medications



Soriatane (acitretin)

Soriatane is an oral retinoid, which is a synthetic form of vitamin A. The FDA approved Soriatane in 1997 as the only oral retinoid for treating psoriasis. Soriatane can be taken daily, and the amount depends on the type and severity of psoriasis.

The exact way Soriatane works to control psoriasis is unknown. In general, retinoids help control the multiplication of cells, including the speed at which skin cells grow and shed. Soriatane differs from other oral systemic treatments for psoriasis in that it does not suppress the immune system. This may be a helpful choice for patients who have a compromised immune system or are using treatments that suppress the immune system.

Soriatane can be safely used with phototherapy to treat psoriasis. Using these treatments together may reduce possible side effects and the frequency of phototherapy treatments. This can be more convenient and affordable for patients. Always talk to your health care provider before combining any treatments.⁸ Due to high risk of birth defects, women that are of childbearing years and who are considering Soriatane should talk with their health care provider about if Soriatane is right for them. **Since Soriatane can stay in your body for up to 3 years, it is important to stop using Soriatane well before you plan on becoming pregnant.** If you do decide to use Soriatane, you will access it through a restricted program called *Do Your P.A.R.T.*²⁰ There is limited data to suggest that Soriatane does not impact fertility or lead to birth defects when men are taking the treatment.⁸

Talk with your health care provider if you have any concerns about the potential risks or side effects of any treatment.

Avoid if you²⁰

- Are pregnant or plan to become pregnant in the next 3 years
- Are sensitive or allergic to retinoids
- Have a high level of fat in the blood that cannot be controlled by medications
- Have liver problems
- Have kidney problems
- Are taking methotrexate or tetracyclines (a type of antibiotic)

Common side effects include²⁰

- Chapped and dry lips
- Dry mouth
- Joint pain
- Tight muscles
- Hair loss
- Dry eyes
- Rise in blood fats (lipids)

Potential risks include²⁰

- Birth defects in the fetus if you are pregnant or plan to become pregnant
- Liver damage

May not be safe to use with²⁰

- Natural sunlight or sunlamps (UV phototherapy may still be safe at the right dose)
- Vitamin A supplements and oral retinoids
- Some other prescription medications



Sotyktu (deucravacitinib)

Sotyktu is a selective, allosteric tyrosine kinase 2 (TYK2) inhibitor. TYK2 is a member of Janus Kinase (JAK) family. This means it targets a specific part of the immune system (TYK2 molecules) to reduce the overactive immune response associated with psoriasis.²¹

The FDA approved Sotyktu in 2022 to treat moderate-to-severe plaque psoriasis in adults who are candidates for systemic treatments or phototherapy. Sotyktu is a pill (6mg) taken daily with or without food.

Talk with your health care provider if you have any concerns about the potential risks or side effects of any treatment.

Avoid if you²¹

- Are allergic to Sotyktu or any ingredients in the medication
- Have recently received or are scheduled to receive a live vaccine
- Are pregnant or planning to become pregnant
- Are breastfeeding or planning to breastfeed
- Have an active tuberculosis (TB) infection or have been in close contact with someone who has TB
- Have or have had hepatitis B or C
- Have an active, serious infection
- Have severe liver impairment (Child-Pugh C)

Common side effects include²¹

- Common cold
- Sore throat and sinus infections (upper respiratory infections)
- Cold sores (herpes simplex)
- Sores on inner lips, gums, tongue, or roof of the mouth (canker sores)
- Inflamed hair pores (folliculitis)
- Acne

Potential risks include²¹

- Serious infections
- Cancer
- Muscle problems (rhabdomyolysis)



Xeljanz and Xeljanz XR (tofacitinib)

The FDA approved the use of Xeljanz and Xeljanz XR to treat psoriatic arthritis in 2017. Xeljanz targets a specific part of the immune system (Janus kinase) to reduce the overactive response associated with psoriatic disease.

Xeljanz is taken by mouth twice daily as a pill. Xeljanz XR is an extended-release formula of the same treatment and is taken by mouth once daily as a pill.

Talk with your health care provider if you have any concerns about the potential risks or side effects of any treatment.

Avoid if you²²

- Have an active infection

Potential risks include²²

- High lipid levels
- High liver enzymes
- Low blood count
- Lymphoma and other malignancies
- Gastrointestinal perforations

Common side effects include²²

- Upper respiratory infection
- Headache
- Diarrhea
- Nasal congestion
- Sore throat
- Runny nose
- High blood pressure

May not be safe to use with²²

- Immunosuppressive drugs such as cyclosporine and tacrolimus
- Other prescription medications

Finding the Right Treatment for You

When you start a new psoriasis treatment, you may have some questions for your provider. How long will it be before a treatment starts to work? How much improvement can you hope to see? At what point should you try something different?

To help answer these questions, the NPF issued a set of treatment targets for people living with psoriasis and their health care providers. These targets were published in November 2016 in the *Journal of the American Academy of Dermatology*. By striving to reach these treatment targets, you may find a treatment that helps you to achieve clear or almost clear skin.

These targets do not recommend a specific kind of treatment since treatments may work differently for different people. The target sets a goal of how much improvement you should expect to see after a certain amount of time.

The goals of Treat to Target

The primary Treat to Target goal is to get your psoriasis down to 1% of body surface area (BSA) or less in 3 to 6 months after starting a new treatment. You can measure your BSA and track your progress on your own. Your entire hand (the palm, fingers, and thumb) is equal to about 1% of your BSA. Learn more about measuring your BSA affected by psoriasis at psoriasis.org/watch-and-listen/measuring-body-surface-area.

You can track how much of your BSA is affected by psoriasis before you start a treatment. Then you can compare that to your BSA affected by psoriasis after being on your treatment plan for 3 to 6 months.

The following are the defined treatment targets for psoriasis according to the Medical Board from the NPF²³:

Time after starting a new treatment plan	Treatment target
3 months	1% or less of your BSA affected by psoriasis*
6 months	1% or less of your BSA affected by psoriasis

*It may be acceptable to have less than 3% of your BSA affected by psoriasis (or have experienced 75% improvement) at this time.

When you have been on a treatment for 3 months, check in with your health care provider to see if your psoriasis has met the treatment target.

- If you have less than 1% BSA affected by psoriasis, you have met the goal. Your health care provider can recommend how to continue to keep your psoriasis at this level.
- If after 3 months, you have experienced an acceptable response (psoriasis covering 3% or less of your BSA or you have experienced 75% improvement), it is important to talk with your health care provider. You and your health care provider may decide to continue with your treatment plan for another 3 months or to try different treatment options.

Once you have reached that 1% goal, you should check in with your provider every 6 months to make sure you are keeping the level of clearance.²³

If you have psoriatic arthritis, your health care provider may also recommend following a treat to target approach.²⁴ The best strategy right now is for you and your provider to discuss goals for your treatment and to determine a realistic timeline to wait for desired results.

Considering different treatment options

If you do not meet the target after 6 months—or if you are not seeing acceptable improvement after 3 months—your provider may suggest a different treatment option. Changing the dose, adding a new treatment to the treatment plan you are already on, or switching to a different treatment are all possibilities. It may take time to find the treatment(s) that works for you, so keep working with your provider until you meet your treatment goal.

Targets can affect overall health

Remember, getting clear skin is a goal worth working toward. Improving your psoriasis can improve your overall health too. Research tells us that psoriasis is associated with other diseases, including psoriatic arthritis, heart disease, and diabetes.²⁵ These health conditions related to psoriasis are called comorbidities. Managing your psoriasis may help to lower your risk for comorbidities. Let these targets guide you on a path to clear skin and better health.

Talk with your health care provider

Now that you are more familiar with your treatment options, talk with your health care provider about which treatment might be right for you. Your provider can help make a recommendation based on the severity of your disease and your health history. If you need help preparing for your appointments or making the most of your health care visits, request our *Take an Active Role in Your Health Care* booklet. Contact the Patient Navigation Center by calling 1-800-723-9166 or emailing education@psoriasis.org.



Thank you to our donors who have made it possible to provide resources like this. You can make a donation at [PSORIASIS.ORG/GIVEPNC](https://psoriasis.org/givepnc).

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Work with a Patient Navigator

The NPF Patient Navigation Center (PNC) provides assistance to anyone impacted by psoriatic disease, including families and caregivers. Information provided by the PNC is for educational use only and does not replace medical care or advice.

You can reach our navigators by email and phone Monday-Friday 7:30 a.m. - 4:30 p.m. PST.

Email: education@psoriasis.org

Call: 1-800-723-9166 (option 1)



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